MIL-I-82356/25
4 April 1969

MILITARY SPECIFICATION

INSTRUMENTS, SIMULATED, AIRCRAFT TRAINING DEVICES; INDICATED AIRSPEED - MACH NUMBER AND MAXIMUM ALLOWABLE AIRSPEED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

The technical requirements for procuring the instrument described herein shall consist of this document and the issue in effect of Specification MIL-I-82356.

1. SCOPE

1.1 This specification covers the detail requirements for simulated indicated airspeed, mach number, and maximum allowable airspeed indicators for use in aircraft training devices.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-5419	Indicator: Airspeed, Pitot Static, Maximum Allowable Airspeed, 50 to 650 Knots
MIL-S-20708	Synchros, General Specification for
MIL-I-21718	Indicator, Airspeed, Pitot Static, Maximum Allowable Airspeed, 40 to 400 Knots
MIL-I-25137	Indicator, Airspeed, Mach Number, And Maximum Allowable Airspeed, Type ME-4
MIL-I-27197	Indicator, Airspeed, Mach Number, And Maximum Allowable Speed AVU-8/A
MIL-I-82356	Instruments, Simulated, For Aircraft Training Devices, General Specification For

FSC 6930

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STANDARDS

Military

MS 25333	Indicator, Airspeed, Pitot Static, Maximum Allowable Airspeed, 40 to 400 Knots
MS 28048	Indicator, Indicated Airspeed Pitot Static Maximum Allowable Speed 50-650 Knots
MS 33556	Case, Aircraft Instrument, 3 1/4 Inch Size, Standard Dimensions For

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Procuring Contracting Officer.)

3. REQUIREMENTS

3.1 General.- The requirements of MIL-I-82356 and this specification shall apply.

3.2 Design basis.- Each simulated indicated airspeed (IAS), Mach number, and maximum allowable airspeed (MAAS) indicator shall conform to the applicable operational instrument specified in Table I of this specification and paragraphs 3.3, 3.4, and 3.5.

3.3 Indicated airspeed activation.- The pointer used for IAS display shall be driven by a 26V-400 hertz receiver synchro which shall be electrically zeroed per MIL-S-20708 with the pointer at 0 knots. The ratio between the synchro shaft and pointer shall be 1:1.

3.4 Mach number dial activation.- The dial used for the mach number display shall be driven by a 26V-400 hertz motor generator with a 100:1 ratio gear box. This shall also drive a 26V-400 hertz control transformer which shall have a ratio of 1:1 with the mach number dial. The motor generator shall be so wired with respect to the power input leads at the connector that its pinion shaft output at the gear box shall rotate counterclockwise when viewed from in front.

The control transformer shall be electrically zeroed with the mach 0.5 number lined up with 330 knots indication on the IAS display.

3.5 Maximum allowable airspeed activation.- The NAAS pointer shall be driven by a 26V-400 hertz receiver synchro through a mechanism which shall produce the proper indication with respect to the IAS and the mach number indications. It shall also be provided with facility for locking the pointer at either 600, 700, or 800 knots for electrically zeroing the driving receiver synchro at any of the three speeds.

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3.6 Airspeed and mach number setting indices.- A hand operated knob identical in appearance and function to the hand operated knob on the operational instrument for adjusting a mach number setting-index and/or an airspeed setting-index shall be provided for each simulated instrument, as applicable and shall function in the exact manner of the operational instrument.

3.7 Wiring.- Wiring shall be in accordance with the schematics of Table I as applicable.

3.8 Case.- With the exception of length, which shall not exceed limits specified in the footnotes, cases shall be in accordance with Table I of this specification, as applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-I-82356 and as specified in 4.2 and 4.2.1.

4.2 Tests.- The following tests shall be performed on each instrument in addition to tests specified in the test requirements of MIL-I-82356.

(a) Backlash of gears

(b) Forward and reverse operation

(c) Running smoothness of gears

(d) No jumping or hesitation of pointers

(e) Smooth operation of index knobs.

4.2.1 Electrical tests.- Electrical tests shall consist of checks on rotating components, electrical zero and running performance.

5. PREPARATION FOR DELIVERY

5.1 The preparation for delivery requirements of MIL-I-82356 shall apply.

6. NOTES

6.1 The notes of MIL-I-82356 shall apply.

CUSTODIANS:
ARMY - AV
NAVY - TD
AIR FORCE - 70

USER ACTIVITIES:
ARMY - AV
NAVY - TD
AIR FORCE - 11, 70

PREPARING ACTIVITY:
NAVY - TD
Project Number 6930-0074

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TABLE I - MOTOR-GENERATOR WITH TWO SYNCHRO RECEIVERS DATA

DESIG-NATION	OPERATIONAL INSTRUMENT BEING SIMULATED	NOMINAL SIZE	*CASE TYPE	SCHEMATIC WIRING PER MIL-I-82356 FIG. NO.
-1	MIL-I-25137	3 inch	MS 33556	11
-2	MIL-I-27197	3 inch	MS 33556	11 **
-3	MIL-I-5419	3 inch	MS 28048	11
-4	MIL-I-21718	3 inch	MS 25333	11

* Maximum case length of simulated instruments may exceed the case length specified in the specifications noted above. However, the maximum case length should not exceed 8 inches without prior approval from the procuring activity.

** Designation denotes an integrally lighted instrument.

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